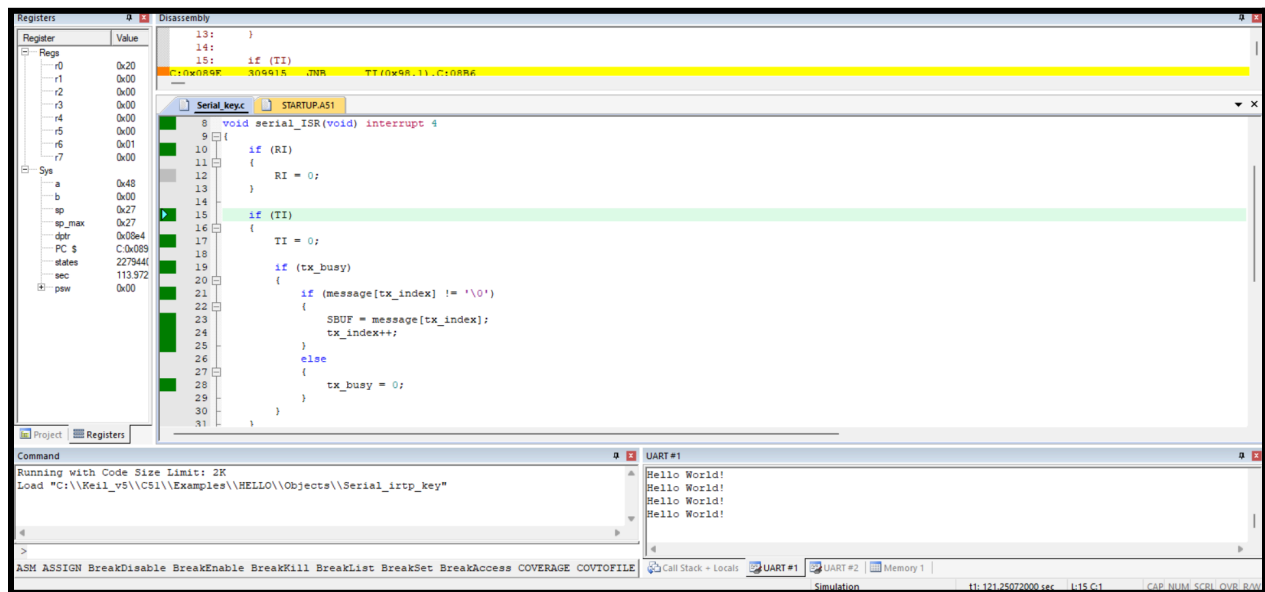
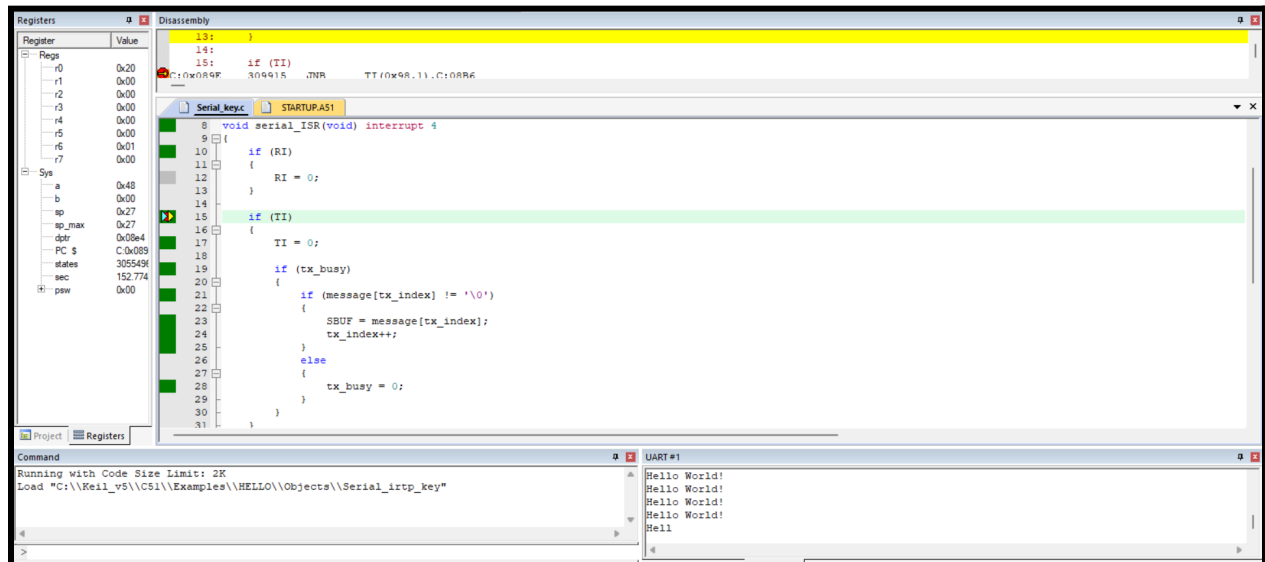


Before Interrupt:



After Interrupt:



C Program:

```
#include <reg51.h>

unsigned char tx_index = 0;
bit tx_busy = 0;

char code message[] = "Hello World!\r\n";

void serial_ISR(void) interrupt 4
```

```

{
    if (RI)
    {
        RI = 0;
    }

    if (TI)
    {
        TI = 0;

        if (tx_busy)
        {
            if (message[tx_index] != '\0')
            {
                SBUF = message[tx_index];
                tx_index++;
            }
            else
            {
                tx_busy = 0;
            }
        }
    }
}

```

```

void main()
{
    TMOD = 0x20;
    TH1 = 0xFD;
    TL1 = 0xFD;

    SCON = 0x50;
    TR1 = 1;

    ES = 1;
    EA = 1;

    while(1)
    {
        if(tx_busy == 0)
        {
            tx_index = 1;
            tx_busy = 1;

            SBUF = message[0];
        }
    }
}

```